School of Computer Science and
Electronic Engineering
University of Essex

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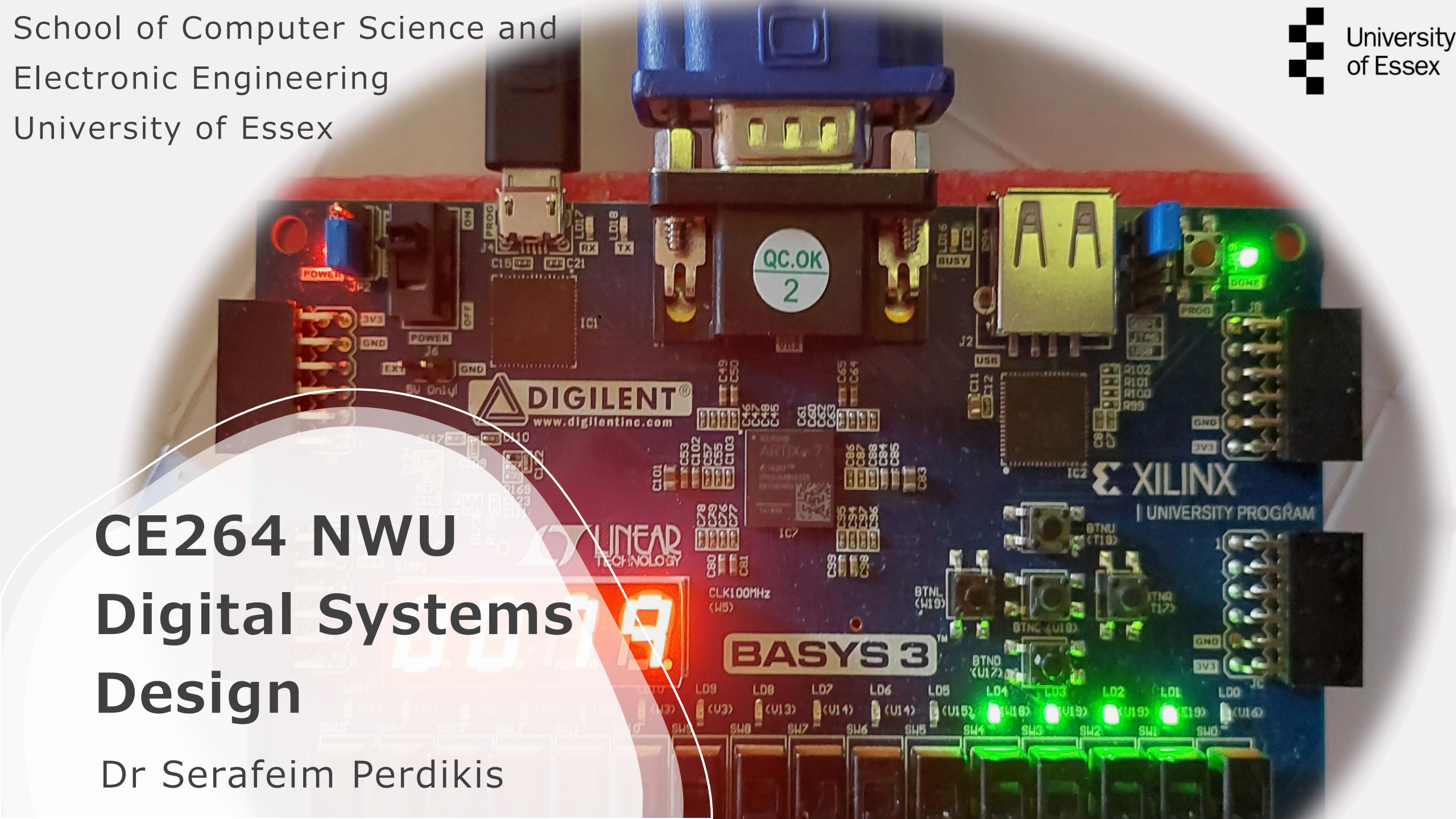

CE264 NWU Digital Systems Design

Dr Serafeim Perdikis

School of Computer Science and
Electronic Engineering
University of Essex

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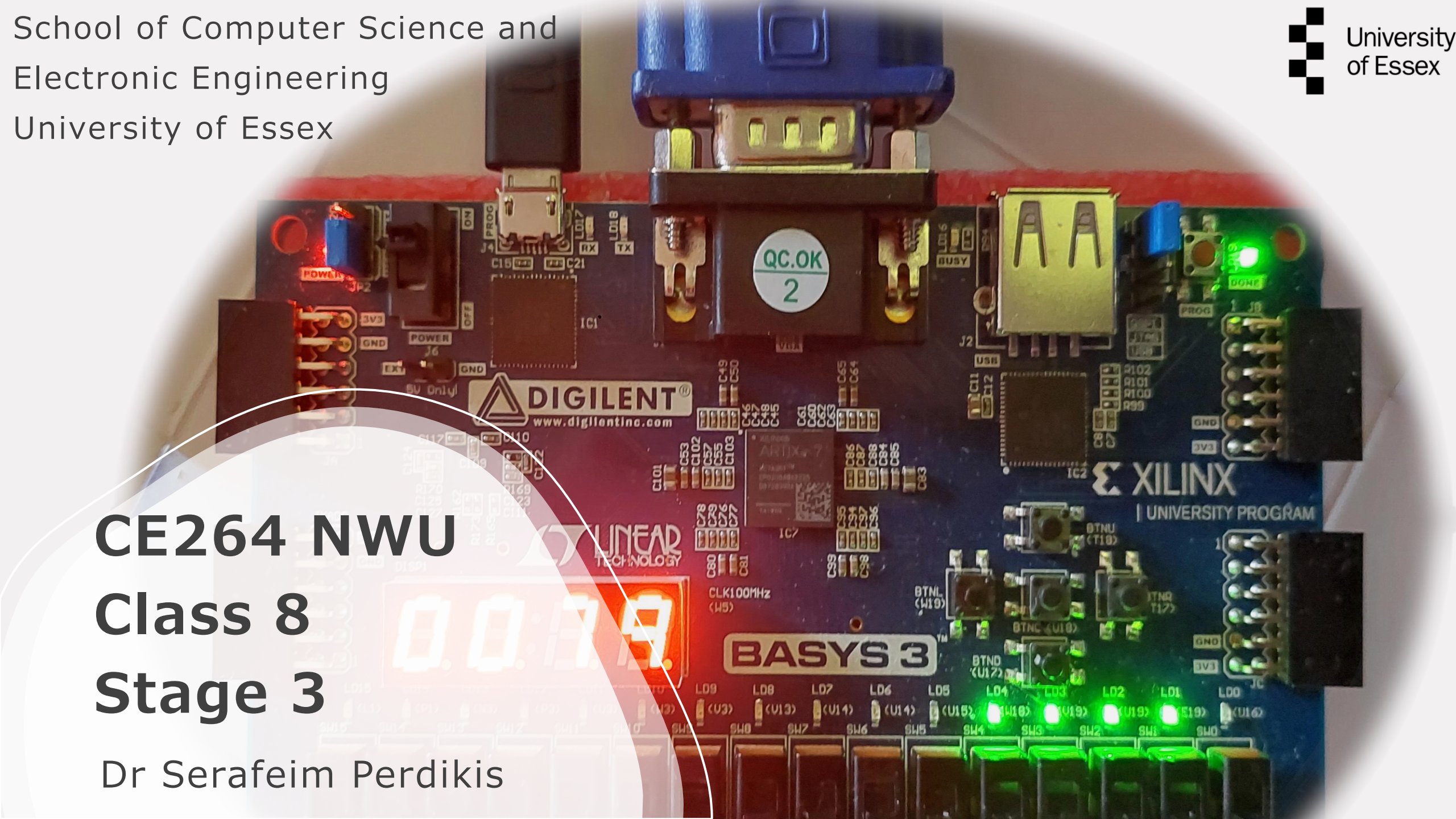
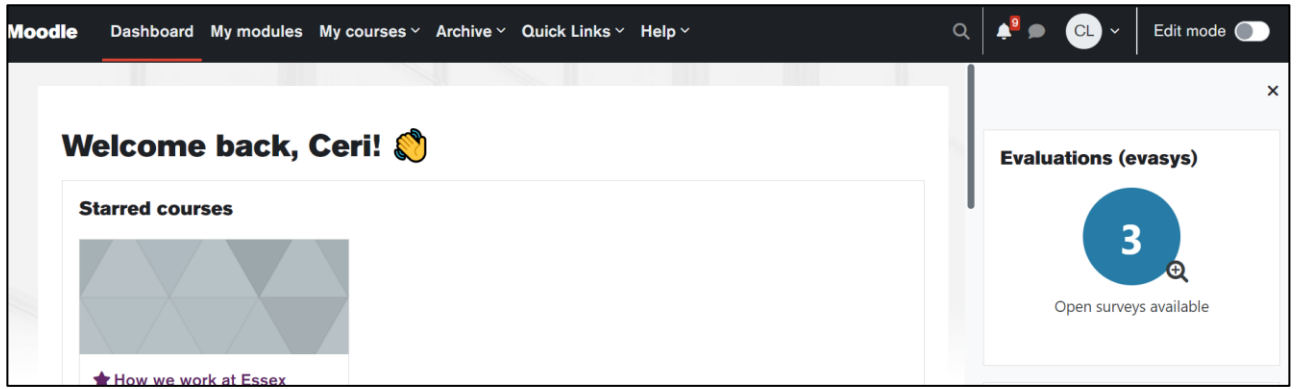


CE264 NWU
Class 8
Stage 3

Dr Serafeim Perdikis

Student Module Feedback

- What's that?
  - Questionnaire for students to evaluate a module
  - Necessary for teachers and TAs to improve delivery!!!
- How do I complete that?
  - You should have received an email to your Essex email address from 'University of Essex surveys' with links to all the surveys available to you.
  - You can also access your available surveys via Moodle:



This will only show if there are surveys available to you.

Lab/material access after April 6th

- Access to the labs (Vivado, Multisim, oscilloscope) and material (Basys 3 board, VGA monitor) will be given after Week 4 and until May 7th (submission day)
- Possibility to:
  - Continue your design with Stage 3 and additional features (open-ended)
  - Test your design and shoot the testing videos required
- Access will be limited so:
  - Focus and push yourselves to finish as much as possible within Week 4
  - Plan well so that you can have designs ready for testing on access day

Resources, collaboration & plagiarism

- Some of you are using designs from earlier year students
  - Yeah, yeah, I know it for a fact, please do not deny it! ☺
- **It is alright, and even recommended, to use resources like:**
  - Designs given to you be colleagues, friends, etc.
  - Designs inspired by your classmates in labs with collaboration
  - Designs you found on the internet (tutorials, forums, etc.)
  - Designs you found in any other written resource (books, etc.)
- …as long as you:
  - use the resource as inspiration only, and submit your own version of it!
  - acknowledge the help you got and cite it in the report!
- **Check Introduction slides for what is, and what is not, plagiarism**

How come it is alright to use such resources?

- Our goal is for you **to learn**
  - Learning is always based on some resources
  - Resources like internet tutorials/forums, books, colleagues in the same team is the normal way we **keep learning as engineers and scientists**
    - **"Learning to learn"** is a major goal of this module (and all other modules)
  - Learning is proven to be more effective and efficient in collaborative learning communities
- We decouple learning from assessment!
  - (Summative) assessment is how you prove to us you have learned
  - …and get a mark

Decoupling learning from assessment

- Measures for reliable assessment:
  - **“Justification”/”Discussion”** and **testing** far more important than schematics in marking criteria
    - Explanations must be in your own words, different than other reports
  - Testing videos must include your face, student ID card and voice!
    - If not, they will not be taken into account!
  - You must also submit your schematics and constraints file
    - Very easy to detect plagiarism!
  - Stage 3 design is novel and depends on your student ID number
- Can you still cheat? Probably…
  - …but you will have learned something
  - …or, get a very bad mark!

Dynamic Programmable Video Test Card (Stage 3)	**55**
Basic feature 1: Draw colored rectangle in arbitrary position with given dimensions	***14***
Schematic implementing feature	2
Justification of the schematics’ function	6
Video evidence of feature functionality	6
Basic feature 2: Move rectangle with buttons obeying screen limits	***12***
Schematic implementing feature	2
Justification of the schematics’ function	5
Video evidence of feature functionality	5

Decoupling learning from assessment

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Refuse to help colleagues that do not play by the rules!

Dynamic Programmable Video Test Card (Stage 3)	**55**
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Justification of the schematics' function	6
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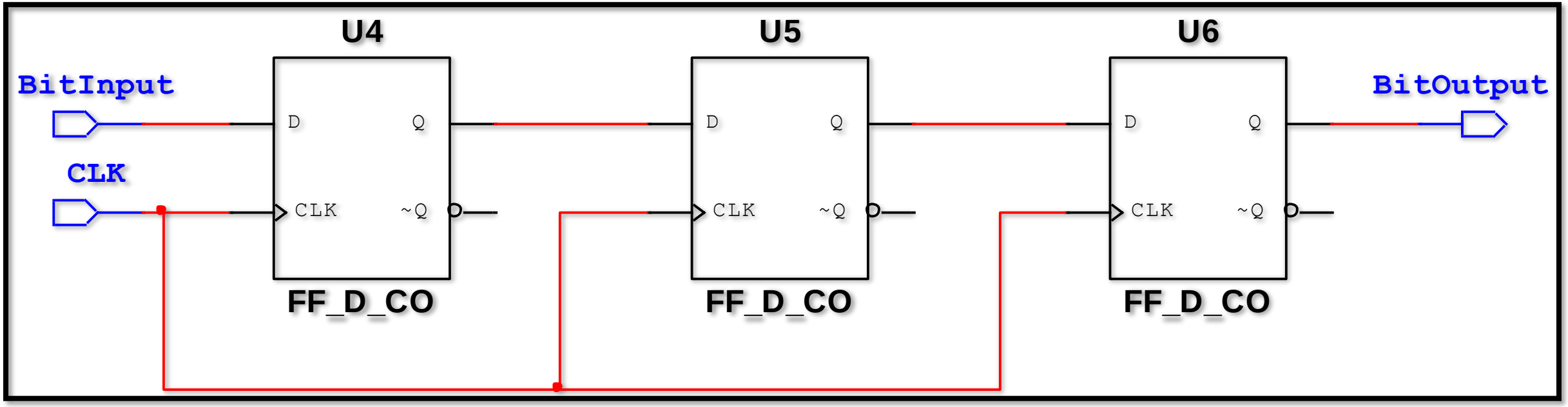
Schematic implementing feature	2
Justification of the schematics' function	5
Video evidence of feature functionality	5

Justification & Discussion of your designs

- “Justification”/”Discussion”
  - Most valuable of marking criteria
  - Used to assess whether your submission is your own work
- So, what is that?
  - Text in your report that **explains** the logic of your design
  - Put it underneath (or, at least, near) the corresponding schematic
- What makes a good justification?
  - Imagine you are trying to help a friend understand your design!
  - **Functionality of each component** present in the design must be clear
    - and how this functionality contributes to the logic of the overall design
  - **Connectivity** between components and with general inputs/outputs
    - Explain how connecting output A of component X to input B of component Y contributes to the logic of your design
    - Do this at least for the main (most important) inputs/outputs if design too big
    - One more reason to use hierarchical design: break down justification to smaller schematics

Justification example: 3-bit LR shift register

- Good justification yields 70-100% of (justification) marks
- Medium (mediocre) justification yields 40-65% of marks
- Bad justification yields 10-35% of marks
- No justification yields 0 marks



Bad justification

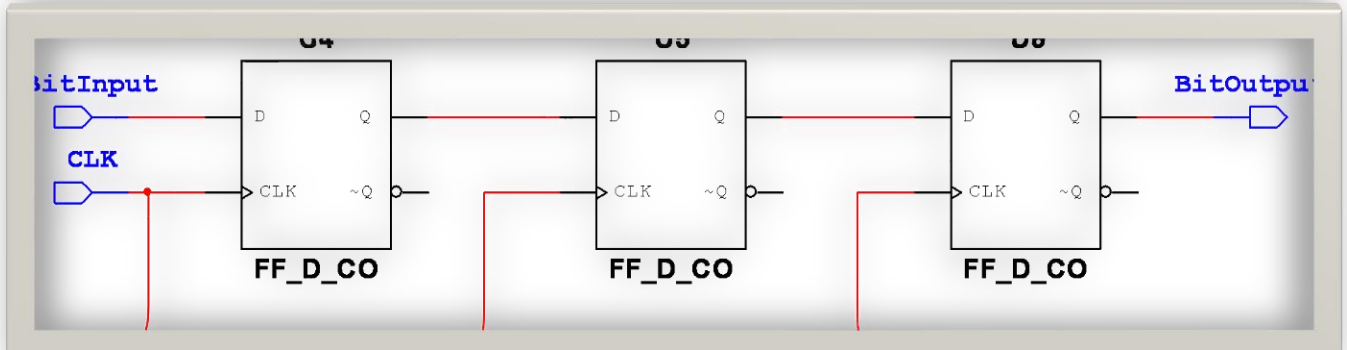
"The following schematic shows my left-right shift register design"

"The following schematic shows my left-right shift register design. This shift register design inputs data serially from left to right."

"The following schematic shows my left-right shift register design. This shift register design consists of D flip-flops and inputs data serially from left to right."

What's wrong?

- No component functionality
- No justification of connectivity

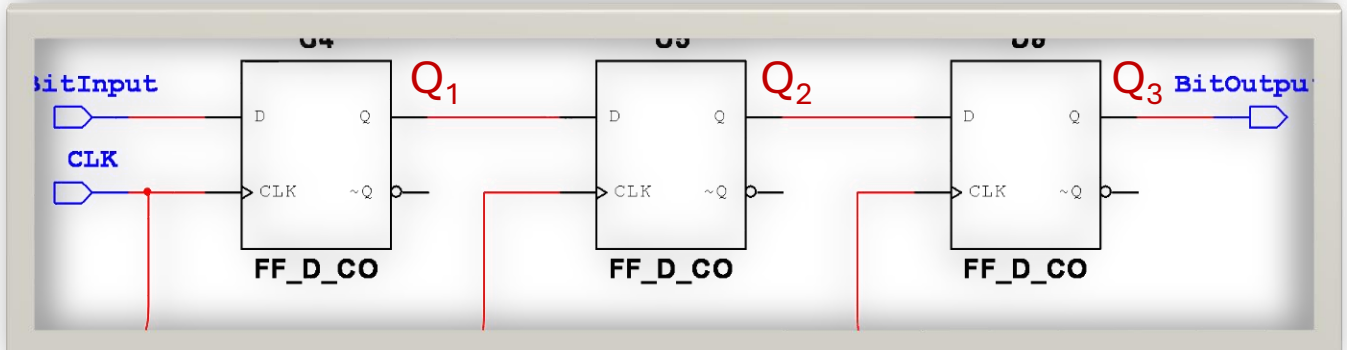


Mediocre justification

"The following schematic shows my left-right shift register design. This shift register design consists of 3 D flip-flops and inputs data serially from left to right. This is because the basic functionality of a D- flip-flop is to forward to its output pin Q, whatever input is currently applied to its input pin D . At each clock, each bit moves by one position to the right, and the right-most bit is lost.

What's wrong?

- No justification of connectivity
- Lacks important detail

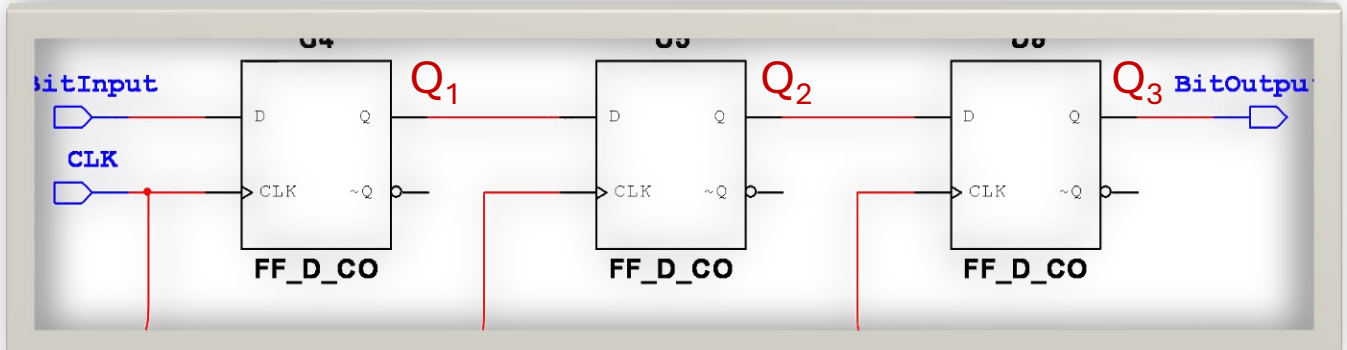


Good justification

"The following schematic shows my left-right shift register design. This shift register design consists of 3 D flip-flops and inputs data serially from left to right. When a new bit is available for input, it is applied to the BitInput pin and appears at the output Q_1 of the first D flip-flop at the next clock. This is because the basic functionality of a D- flip-flop is to forward to its output pin Q, whatever input (0 or 1) is currently applied to its input pin D . At each clock, each bit moves by one position to the right, and the right-most bit is lost (ejected from the register). This is because the output Q of each flip-flop is connected to the input D of the next flip-flop on its right side. Q3 (the oldest bit stored) is exposed through the BitOutput to the rest of the circuit."

What's wrong?

- Nothing. All basic info is there.
- Could have some more detail…

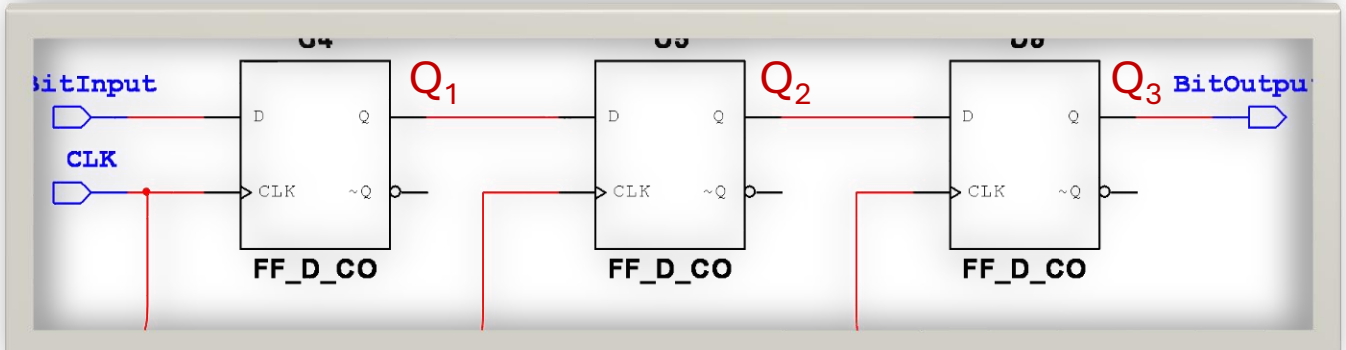


Excellent justification

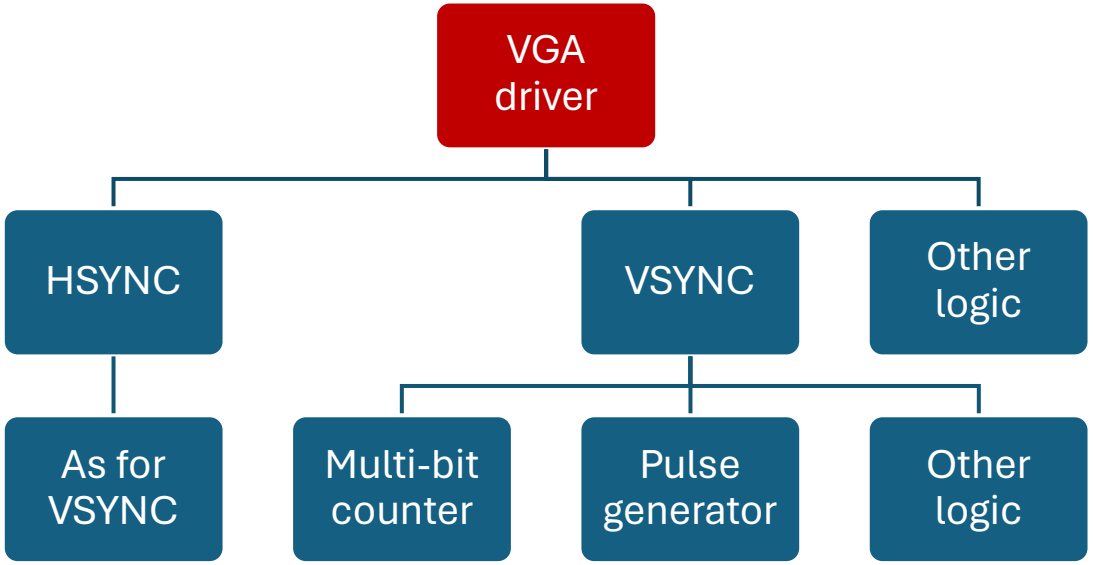
"The following schematic shows my left-right shift register design. This shift register design consists of 3 D flip-flops and inputs data serially from left to right. The output of each flip-flop holds 1 bit of memory, so that the overall register holds 3 bits of memory: the left Q_1, middle Q_2 and the right bit Q_3. The bits of memory are inputted one-by-one into the register through the BitInput input. When a new bit is available for input, it is applied to the BitInput pin and appears at the output Q_1 of the first D flip-flop at the next clock. This is because the basic functionality of a D- flip-flop is to forward to its output pin Q, whatever input (0 or 1) is currently applied to its input pin D . At each clock, not only the current BitInput is transferred to Q_1, but simultaneously also i) the memory bit Q_1 is transferred to Q_2 (since Q_1 was connected to the D input of the second flip-flop), ii) the memory bit Q_2 is transferred to Q_3, and the memory bit Q_3 is lost (ejected from the register). This is because the output Q of each flip-flop is connected to the input D of the next flip-flop on its right side. Hence, at each clock, the memory bits of the left-right shift register move by one position to the right, the right-most bit (the third bit Q_3 in this case) is ejected and lost, and a new bit of information provided by BitInput is stored to the left-most bit Q_1. The transitions are synchronous (concurrent) because the clock input of all 3 D flip-flops is connected to the same input clock of the overall shift-register, CLK. Since only Q3 is exposed through the BitOutput, only the left-most (oldest) bit of this 3-bit memory can be read out at each moment."

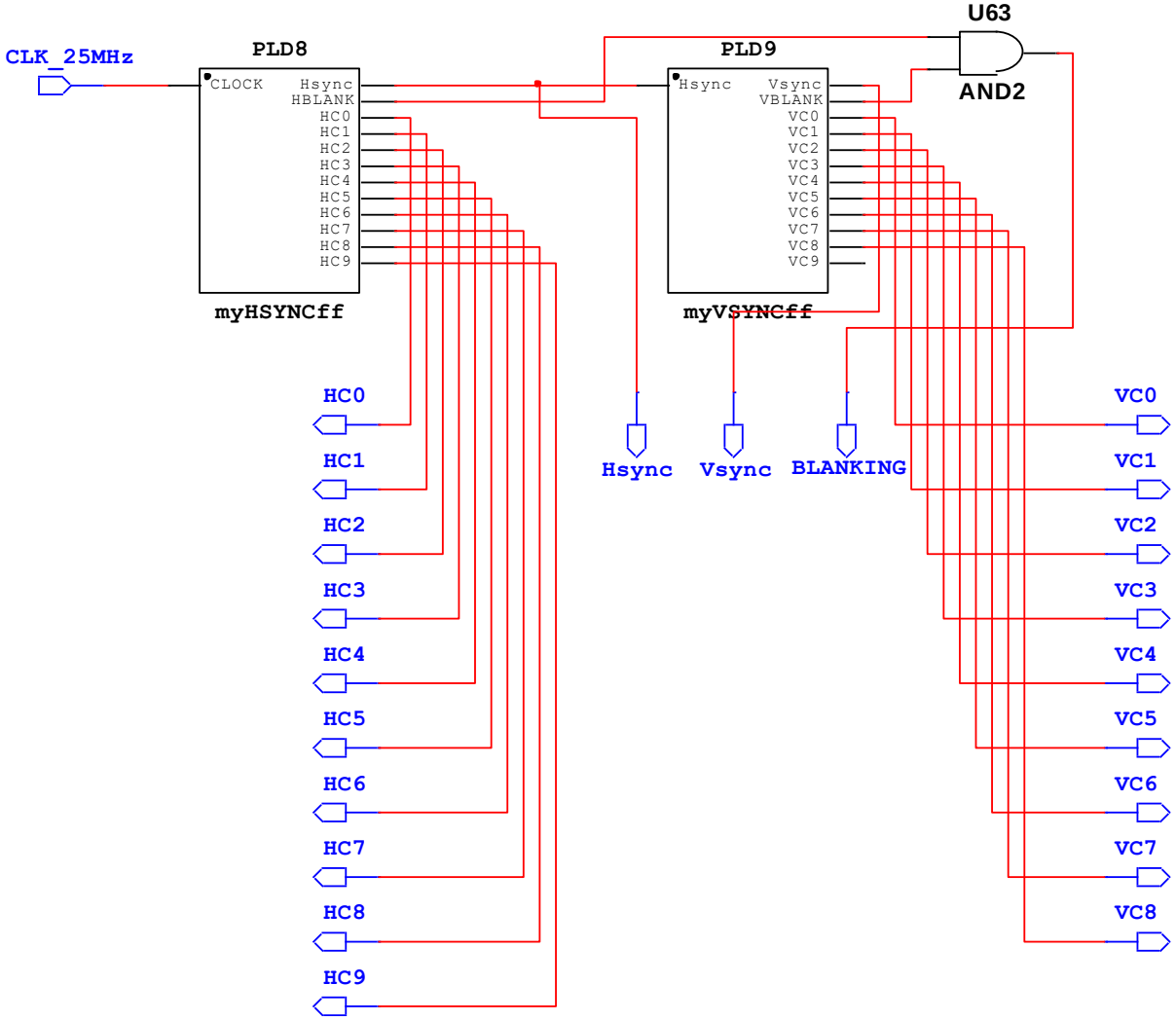
What's wrong?

- Nothing! All necessary info is there!
- Very convincing
- Could also add an example



Example of good hierarchical design





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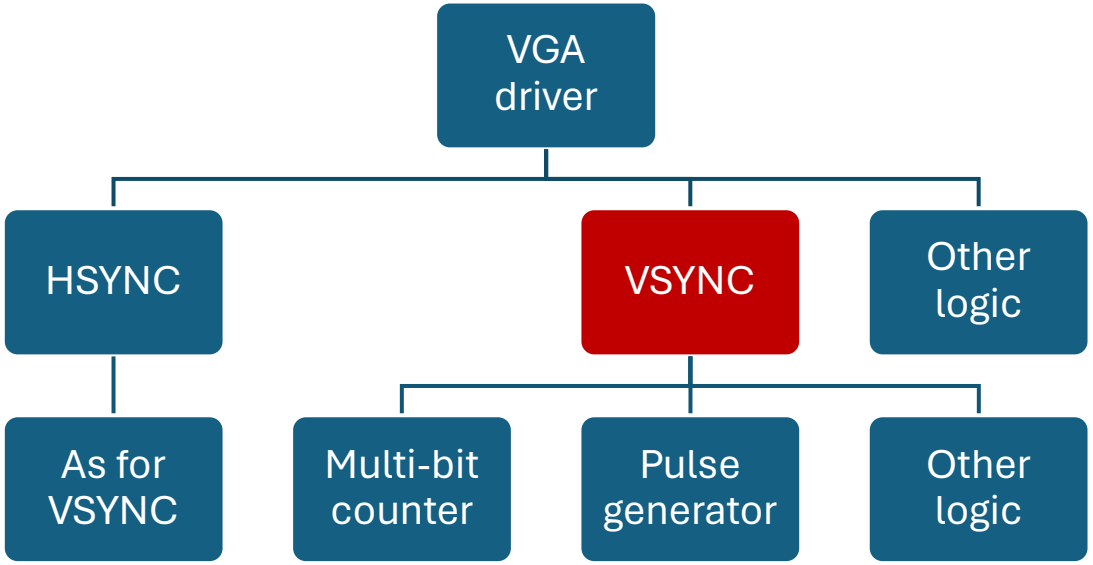
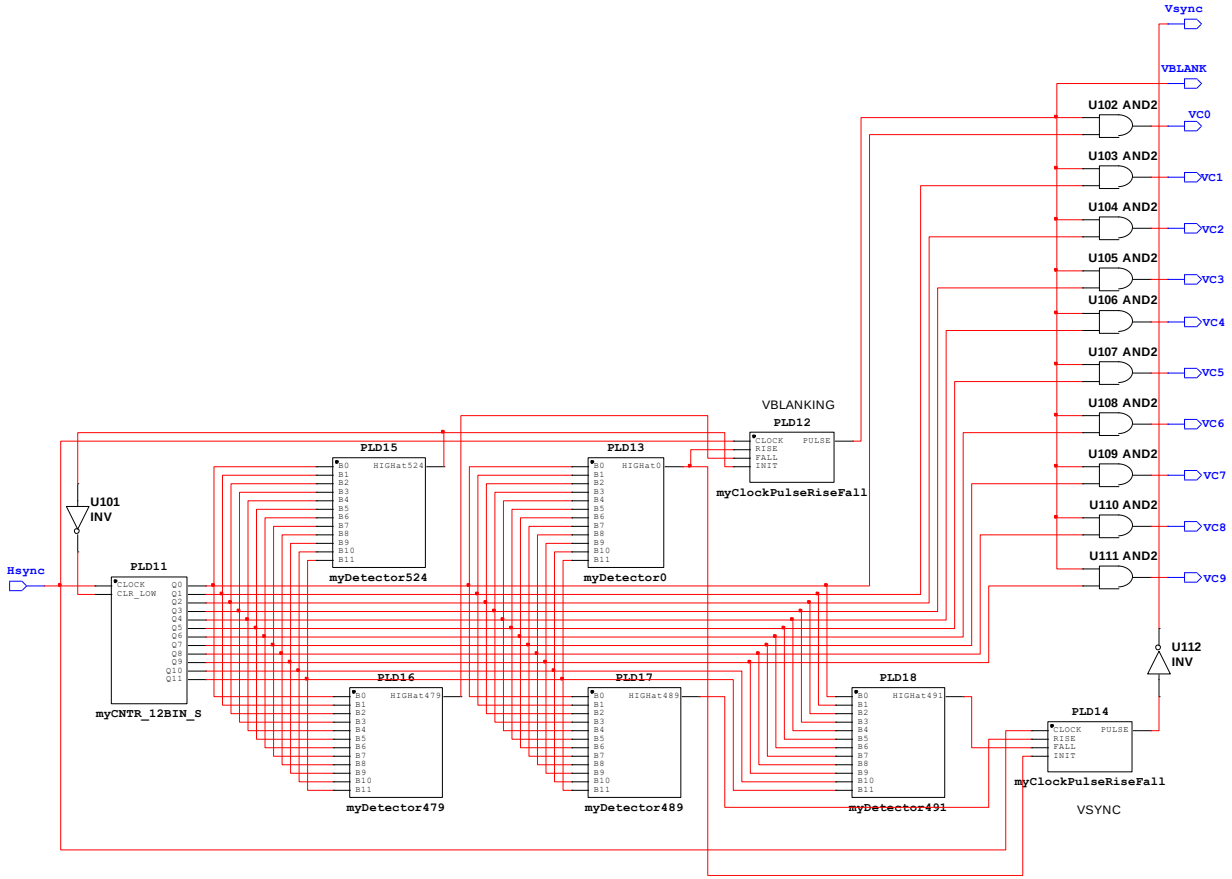
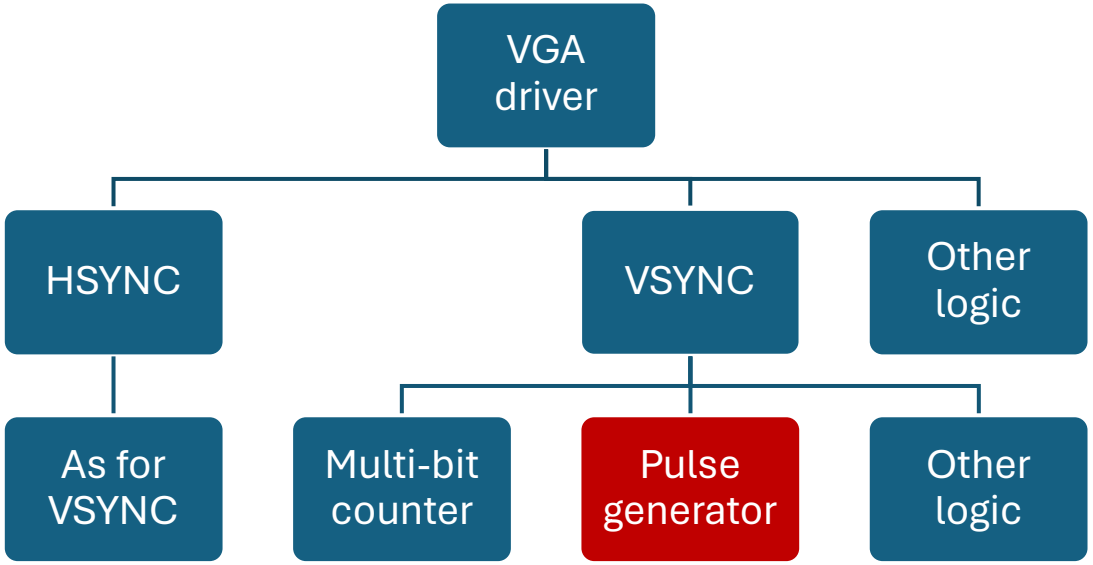




Example of good hierarchical design



?

Try to figure this out yourselves!

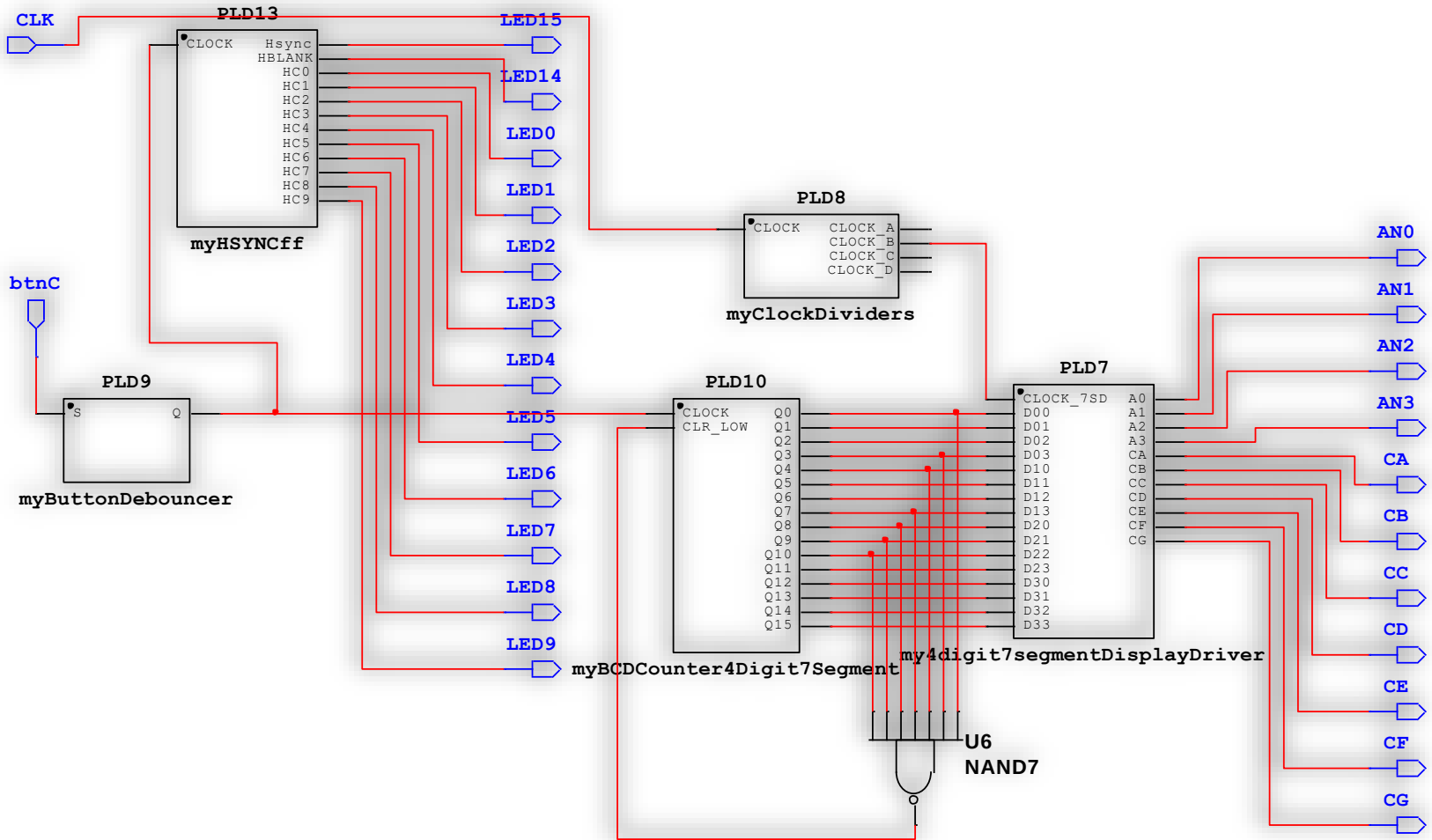
Example of unit-testing HSYNC

How to be sure our HSYNC works?

- Implement the whole VGA driver and test with the monitor? ☒
- Simulate the circuit in Multsim? ☒
- Simulate the circuit in Vivado? ☒
- Check the signal with the oscilloscope? ☑
- Make a unit-test wrapper circuit and test on the board? ☑

Example of unit-testing HSYNC

Submit ALL unit-testing schematics to FASER with their Constraints.xdc



Stage 3 design

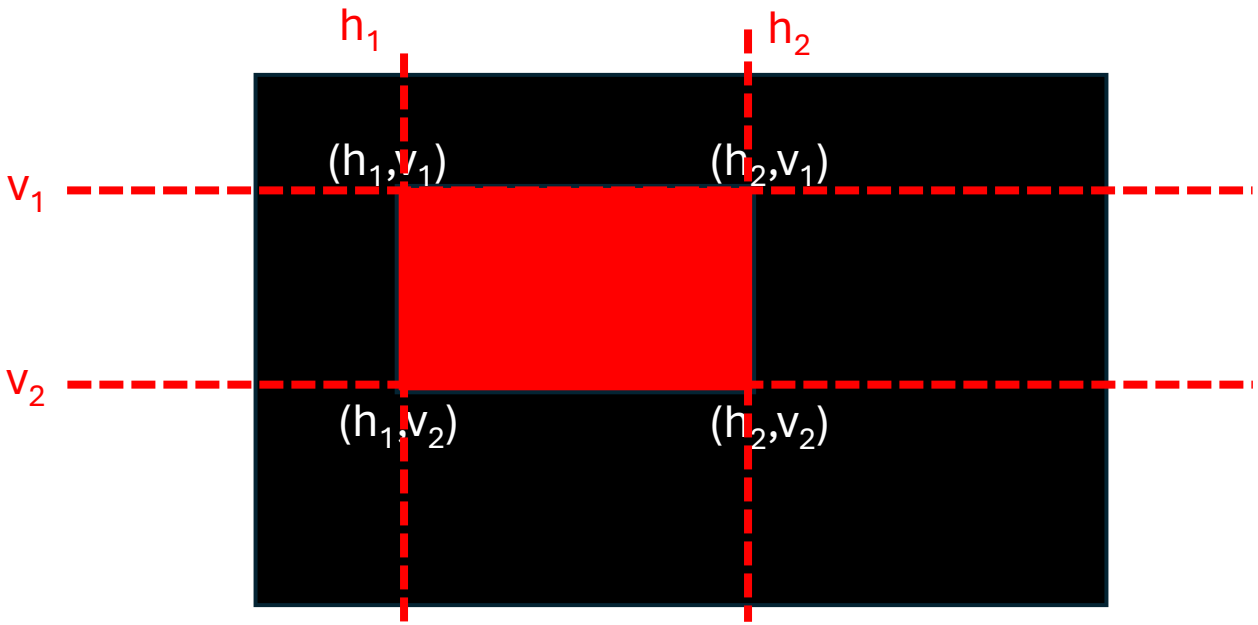
- Moveable, coloured rectangle on a black background
- Five main features:
  - Draw rectangle on arbitrary position (of the top-left corner) on the screen
    - Rectangle dimensions W(idth) x H(eight) can be hardcoded
    - Rectangle dimensions W x H are derived from your student ID
  - Move rectangle with buttons obeying screen limits
    - All directions: up, down, left, right
    - The rectangle should not go out of the screen
  - Display rectangle coordinates with 7-segment display and LEDs
  - Store two colors in memory with switches and button press
  - Rectangle colouring modes 1 and 2 controlled by a switch
    - Change color with button press
    - Automatic flicker from one stored colour to the other

Video demo of Stage 3 design features

- https://panopto.essex.ac.uk/Panopto/Pages/Viewer.aspx?id=85242e6e-ccca-4a21-b30c-b13e00cad544

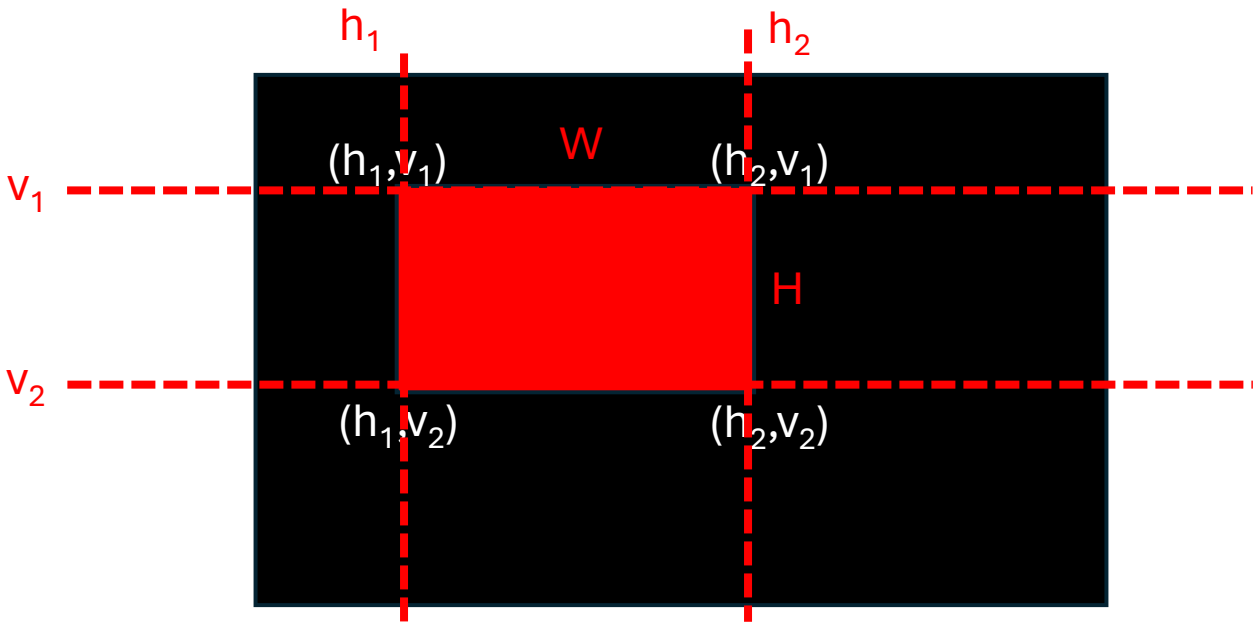
Design hints: drawing a rectangle

- Basic idea:
  - Some color when
    $h_1 < HCOUNT < h_2$ and
    $v_1 < VCOUNT < v_2$
  - Black otherwise
- We have mathematical "comparisons"…
  - …which digital component you think will be useful to implement these?



Design hints: moving the rectangle

- h_1, h_2, v_1, v_2 must now change on button press
  - Which component could do this?
  - You have used this component a lot; do you need the same type, though?
- Dimensions W, H must stay the same!
  - $h_2 = h_1 + W$
  - $v_2 = v_1 + H$
  - What math operation do you need?
  - Is there a component for that?
  - Do you need to cascade?
    - And how exactly?
- Never forget hierarchy!



Design hints: storing colours in memory

- What is the prototypical component for a 1-bit memory?
- How many bits of memory do you need in Basys 3 for each colour you need to store?
- Is there an available component for multi-bit memory, too?
  - If yes, do you need to cascade to get the amount of bits you need?
- You need to forward the switch inputs SWx either to memory 1 or to memory 2
  - What kind of component can forward an input to different outputs based on a select input?
  - The guidelines do not ask for mutual exclusiveness of color storage
    - Meaning, it is in principle possible to store the current colour to both memories
    - Can you also do the storage even more simple then?
  - All the designs that achieve the storage behaviour are acceptable!

Design hints: colour modes

- A bunch of selections:
  - Select colour mode according to a switch (0/1 input)
  - Colour mode 1: select colour after button press to toggle colour state
    - Which component toggles a binary state?
  - Colour mode 2: select colour according to clock state
- Which component can select among different inputs which one to forward to a single output, according to one or more select inputs?
  - You have used this logic somewhere already, in Stage 1!
- Check videos in Moodle to understand desired behaviours

Design hints: displaying coordinates

- That sounds easy but…
  - 7-segment display requires coordinates in BCD
  - LED outputs can only show coordinates in binary
- How can you have the coordinates in the correct number system?
  - Is it possible to get each coordinate in the desired format from start?
  - Is it possible to get each coordinate in both formats from start?
  - Is it possible to convert from one system to the other?
    - What components do these kind of conversions?
      - Do we have suitable ones available for these conversions?
    - Can I convert with an algorithmic design?
      - Learn about “Double Dabble”! A VHDL or Verilog implementation will be accepted!
  - More than one designs possible, but are all equally easy to implement?

Design hints: hierarchical design in Stage 3

- That's my own Stage 3 design
  - Is it small and simple?
- How would it look like without hierarchy?
- Would you be able to debug this without hierarchy and unit testing?
- Would you be able to recover from accidental deletions?
- Would you be able to understand it after 1 week?

Thank you!

- Stage 3 should be unguided
  - I wont help you much beyond these hints
- Good luck!!!
- Questions?